Computational and structural bounds for monochromatic boxes in three-dimensional grids

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Abstract

Let $N(c,d)$ denote the least R such that every c-colouring of the grid $[R]^d$ contains a monochromatic *box*, i.e. a set $\{x_1,x_1'\}\times\cdots\times\{x_d,x_d'\}$ with $x_i\neq x_i'$ for every i, all of whose 2^d points share one colour. These numbers — the cube case of the grid Ramsey problem for boxes — satisfy $N(c,d)=\Theta(c^{2^d})$ with the exponent known only up to a factor of d, and almost nothing is known about their exact values. This note is a computational study of the case $d=3$, $c\le 5$.

Our main contribution is computational: *explicit* box-free colourings of $[n]^3$ with $n=23,40,44$ for $c=3,4,5$, each verified by exhaustive search over all $\binom{n}{2}^3$ boxes, which give $N(3,3)\ge 24$, $N(4,3)\ge 41$, $N(5,3)\ge 45$. Cooper–Fenner–Purewal give the matching upper bounds $93,277,657$ for these c but tabulate no cube colourings; we found no lower bounds for $c\ge 3$ in the literature.

For $c=2$ we sharpen the upper bound. Writing $\varphi_c(n)$ for the least number of monochromatic rectangles in a c-colouring of $[n]^2$, the number of features that a slice must use gives the necessary condition $n\varphi_c(n)\le c\binom{n}{2}^2$ (Lemma 15 of Cooper–Fenner–Purewal in packing language). Two exact levels of convexity give $\varphi_2(21)\ge 4205$, so a box-free 2-colouring of $[21]^3$ would need $21\cdot 4205=88305$ features out of the 88200 available; hence $N(2,3)\le 21$, improving the bound 22 obtained from their double-counting recursion. The wall at $n=21$ is the projective plane of order 4: at $n=21$ the continuous form of the argument is exactly tight, and the improvement is purely an integrality effect (Remark 15).

On the structural side we give a layer (feature-packing) formulation, an exact criterion for shift-orbit families, a dimension-raising lemma, and an exact reduction of the three-dimensional problem, for colourings constant on the level sets of $x+y$, to a two-dimensional *corner* condition (Section 4.1). We are explicit throughout about which statements are re-derivations of known results. Code, colourings and verification scripts are available at `https://github.com/dimkadimon/Ramsey`.

1 Introduction

1.1 The problem

For integers $n,d\ge 1$ let $[n]^d=\{1,\dots,n\}^d$. A *d-box* is a set

$$B=\{x_1,y_1\}\times\cdots\times\{x_d,y_d\}\subseteq[n]^d,\qquad x_i\neq y_i \text{ for all } i.$$

(These are the "combinatorial boxes"; geometrically they are the vertex sets of axis-parallel rectangular boxes, i.e. of prisms when $d=3$.) A colouring $f\colon[n]^d\to[c]$ is *box-free*, and $[n]^d$ is *c-colourable*, if some colour class restriction has no monochromatic box; otherwise $[n]^d$ is *c-guaranteed*. Finally

$$N(c,d)=\min\{R:[R]^d \text{ is } c\text{-guaranteed}\}.$$

We write $N(c,d)[3]$ for the three-dimensional case, which is the subject of this note. Note that c-guaranteedness is monotone: if $[n]^d$ is c-guaranteed then so is $[n']^d$ for $n'\ge n$, and a box-free colouring of $[n']^d$ restricts to one of $[n]^d$ for $n\le n'$. The latter is used constantly below.

The problem is the cube case of the grid Ramsey problem for boxes. The general study is due to Cooper, Fenner and Purewal [2], who determine the order of growth of the largest volume of

a c-colourable grid, bound the volume of minimal obstructions, and analyse the case $c = 2, d = 3$ computationally; closely related estimates for equilateral grids go back to Agnarsson, Doerr and Schoen [1]. The problem also appears as the Ramsey problem for hyperoctahedra $K_d(2)$ in complete d-partite d-uniform hypergraphs, and relaxing van der Waerden's theorem to boxes is one motivation [2, 3].

For $d = 1$ the answer is immediate: $[n]$ is c-guaranteed iff $n \geq c + 1$, so $N(c, 1) = c + 1$. For general d the known bounds are

$$2^{2^{d/(2d)}} \lesssim N(c, d) \lesssim (d + 2)\, 2^{2^{d-1}}, \tag{1}$$

see [2, Thm. 2] for the lower bound and [2, Cor. 3.6] for the upper bound; both are proved by counting monochromatic boxes. The exponent is thus known only up to a factor $\Theta(d)$, and closing this gap appears to be open.

1.2 What this note does, and what it does not

This note is an extended computational and structural study of the case $d = 3$ with $c \leq 5$. Our contribution is deliberately stated in two parts.

What is (as far as we can determine) not in the literature.

(i) **Lower bounds for $c \geq 3$.** Explicit box-free colourings of $[n]^3$ for $c = 3, 4, 5$ with $n = 23, 40, 44$, each verified exhaustively (Section 3); these give $N(3, 3) \geq 24$, $N(4, 3) \geq 41$, $N(5, 3) \geq 45$. [2] tabulates no cube colourings for $c \geq 3$, and we found no lower bounds for these parameters elsewhere. These are the most substantial new numerical statements in this note.

(ii) A layer formulation (Proposition 2) that converts the problem into a packing problem for monochromatic rectangles, together with measurements of how much of the "feature space" optimal colourings actually use (Table 2).

(iii) An exact criterion for shift-orbit constructions (Theorem 4), including the role of rectangles fixed by a non-trivial shift; this yields a proof that the single-coordinate family fails for all $n \geq 12$ (Proposition 6) and explains the empirically observed barrier at $n = 12$.

(iv) A dimension-raising lemma (Theorem 7) giving $N(2, d + 1) \leq N(2, d)$, hence $N(2, d) \geq 12$ for all $d \geq 3$, constructively.

(v) **An exact one-dimensional-shift reduction** (Proposition 9): for colourings of the form $\chi(x, y, z) = f(x + y, z)$ the three-dimensional condition is *equivalent* to a family of two-dimensional corner conditions on the level sets of $x + y$, which replaces n^3 free bits by $2n^2$ and puts the sum-type algebraic colourings into a common framework.

(vi) **An improved upper bound for the two-colour cube**: $N(2, 3) \leq 21$ (Theorem 14), against the value 22 obtained from Lemma 4 of [2]. The packing condition $n\varphi_c(n) \leq c\binom{n}{2}^2$ is Lemma 15 of [2]; what is new is bounding $\varphi_2(21)$ by two exact levels of convexity and using the integrality of the row composition, which is what makes $n = 21$ fail (Remark 15). The same computation gives nothing new for $c \geq 3$.

What is not new, and is stated here for completeness and comparison.

(i) The Zarankiewicz bound $N(c, d)[3] \leq 22, 93, 277, 657$ $(c = 2, 3, 4, 5)$ coincides *exactly* with the values produced by Lemma 4 of [2] applied to the cube. Our proof (Section 5) is structurally different — a Zarankiewicz/C_4 argument — but it is not stronger, and we make no priority claim. Only the $c = 2$ value is improved, and only by one, in Theorem 14.

(ii) Our $c = 2$ lower bound $N(2, 3) \geq 12$ is weaker than the bound $N(2, 3) \geq 13$ already implied by [2, Fig. 3]: that table asserts that $[12, 12, 27]$ is 2-colourable, and such a colouring restricts to $[12]^3$.

(iii) The random-colouring lower bound of Section 6 is the standard Lovász-local-lemma/first-moment argument of [2, 1].

Accordingly the note should be read as an experimental-mathematics contribution: the numerical content is the lower bounds for $c = 3, 4, 5$ (which are new) together with a one-unit improvement of the $c = 2$ upper bound, and the structural content is the collection of criteria, each of which converts a three-dimensional statement into a lower-dimensional one and thereby identifies where the difficulty lies.

2 A layer formulation

2.1 Features

Fix c and n. For $x \in [n]$ let f_x denote the x-th *slice* of a colouring $f\colon [n]^3 \to [c]$, i.e. the map $(y, z) \mapsto f(x, y, z)$. A rectangle is a set $R = \{y_1, y_2\} \times \{z_1, z_2\} \subseteq [n]^2$ with $y_1 \neq y_2$, $z_1 \neq z_2$, and it is γ-*monochromatic* for f_x if f_x is constantly γ on R.

Definition 1. A *feature* is a pair (γ, R) where $\gamma \in [c]$ and R is a rectangle. A slice f_x *uses* (γ, R) if R is γ-monochromatic for f_x.

Proposition 2 (Packing criterion)**.** *$f\colon [n]^3 \to [c]$ has a monochromatic box if and only if some feature is used by two distinct slices. Consequently $[n]^3$ is c-colourable if and only if there are n colourings $f_1, \ldots, f_n$ of $[n]^2$ whose used-feature sets are pairwise disjoint.*

Proof. A box $\{x_1, x_2\} \times \{y_1, y_2\} \times \{z_1, z_2\}$ is monochromatic in colour γ exactly when both slices f_{x_1} and f_{x_2} are γ-monochromatic on the rectangle $R = \{y_1, y_2\} \times \{z_1, z_2\}$, i.e. when (γ, R) is used by both. □

There are $F_c(n) = c\binom{n}{2}^2$ features in total, and Proposition 2 turns the problem into a set-packing problem. Its value in practice is that the packing condition is checkable in $O(c\, n^4)$ time and that each slice is a bounded, SAT-checkable object.

Corollary 3 (Counting necessary condition)**.** *If $[n]^3$ is c-colourable and every c-colouring of $[n]^2$ uses at least $\nu_c(n)$ features, then $n\, \nu_c(n) \leq F_c(n) = c\binom{n}{2}^2$.*

For $c = 2$ and $n = 12$ this reads $\nu_2(12) \leq 726$, and $\nu_2(12) \approx 424$ in practice, so the counting condition is far from tight: the obstruction is geometric, not numerical.

3 Constructions

3.1 Algorithm

Slices are built one at a time. Given the multiset of features already used by slices $1, \ldots, x - 1$, we ask a SAT solver for *any* c-colouring of $[n]^2$ that uses none of them (each feature is a four-literal clause), and then improve that slice by a bit-mask local search which reduces its number of monochromatic rectangles without creating a forbidden feature. The improved slice is committed, its features are added to the forbidden set, and the process repeats. If a slice cannot be found we backtrack or restart. All colourings reported were then verified independently.

c	largest n constructed	conclusion	verification
2	11	$N(2,3) \geq 12$	all 166 375 boxes checked
3	23	$N(3,3) \geq 24$	all 16 194 277 boxes checked
4	40	$N(4,3) \geq 41$	all 474 552 000 boxes checked
5	44	$N(5,3) \geq 45$	all 846 590 536 boxes checked

Table 1: Explicit box-free colourings obtained. Each colouring is stored in the accompanying repository as `results/K{c}_N{n}.json`.

3.2 Results

Two independent verifiers were used: a brute-force enumeration of all $\binom{n}{2}^3$ boxes, and a slice-pair comparison implementing Proposition 2 directly. They agree on all stored colourings and on random test inputs, and both reject deliberately corrupted grids.

For $c = 2$ the construction was in fact found not by search but by an algebraic argument: taking $T_y \subseteq \mathbb{Z}_{11}$ to be the y-th translate of the quadratic residues $\{1, 3, 4, 5, 9\}$ mod 11 (the $(11, 5, 2)$ difference set) and setting

$$f(x, y, z) = 1 \iff (z - x) \bmod 11 \in T_y, \tag{2}$$

gives a box-free 2-colouring of $[11]^3$; see Appendix A. The analytic reason is isolated in Proposition 6 below.

3.3 Feature utilisation

The following table records what fraction of the feature space the colourings of Table 1 consume. For $c \geq 4$ the fraction is small and grows slowly, which suggests that the computational results for large c are limited by machine time rather than by the problem's structure.

c	n	features used	fraction of $F_c(n)$
2	10	1533	37.9%
3	20	21 869	20.2%
4	40	238 271	9.8%
5	44	156 340	3.5%

Table 2: Feature utilisation of the best colourings found.

4 Structural limits of shift-orbit constructions

A natural attempt is to take one base colouring f of $[n]^2$ and let the slices be the translates of f along a cyclic group. Write $v = (a, b) \in (\mathbb{Z}_n)^2$ for a shift vector of order n and set

$$f_x(y, z) = f\big((y - xa) \bmod n,\ (z - xb) \bmod n\big). \tag{3}$$

The single-coordinate family is $v = (0, 1)$; the diagonal family is $v = (1, 1)$; all other slopes are interpolations.

Theorem 4 (Exact criterion)**.** *The colouring* (3) *is box-free if and only if*

(a) no rectangle that is fixed by a non-zero element of $\langle v \rangle$ is monochromatic in f, and

(b) among the rectangles not fixed by any non-zero element of $\langle v \rangle$, at most one per $\langle v \rangle$-orbit is monochromatic in f.

Proof. A box over slices $x_1 < x_2$ is monochromatic in colour γ exactly when the rectangle $R_1 = R - x_1 v$ and $R_2 = R - x_2 v$ are both γ-monochromatic in f, where R is the projected rectangle. Thus we need: for every orbit $\mathcal{O}$ and every γ, the number of γ-monochromatic rectangles in $\mathcal{O}$ is at most the number of slices in which a given rectangle appears. A rectangle R appears in slice x whenever $R - xv$ is monochromatic, i.e. whenever some translate of R is monochromatic, so a rectangle appears in as many slices as the size of its orbit. If the orbit has size n exactly one monochromatic rectangle is allowed; if the orbit is smaller (equivalently R is fixed by some non-zero shift) each element of the orbit appears in more than one slice and no monochromatic rectangle at all is allowed. □

Remark 5. Condition (a) is the non-obvious half. For even n and $v = (1, 1)$, the shift $t = n/2$ acts on the coordinates as $y \mapsto y + n/2$, $z \mapsto z + n/2$, which maps a rectangle with both side lengths exactly $n/2$ to itself. Hence for $n = 12$ a base colouring satisfying (3) must contain *no* monochromatic rectangle spanning exactly 6 rows or exactly 6 columns. Ignoring this condition produces invalid colourings; we mention it because an early version of our local search did exactly that.

Proposition 6 (The single-coordinate family)**.** *For $v = (0, 1)$ the criterion of Theorem 4 is equivalent to: for every colour γ and every pair of columns $\{z_1, z_2\}$, the set $S(\gamma, \{z_1, z_2\}) = \{y : f(y, z_1) = f(y, z_2) = \gamma\}$ is a Sidon set in $\mathbb{Z}_n$ (all pairwise differences distinct up to sign, and — for even n — with no antipodal pair). Consequently this family gives no box-free 2-colouring of $[n]^3$ for any $n \geq 12$.*

Proof. With $d = x_2 - x_1$ and rows $a = y_1 - x_1$, $b = y_2 - x_1$, a monochromatic box requires $a, b, a - d, b - d \in S$. Two two-element sets are translates of one another exactly when they have the same oriented difference, so a repeat of any difference produces a box. The Sidon condition gives $|S|(|S| - 1) \leq n - 1$; a balanced 2-colouring has $|S| \approx n/4$ on average, so $n^2/16 \lesssim n$ is needed, which fails for $n \geq 12$ (and a SAT check confirms unsatisfiability at $n = 12$ in under three minutes). □

For the remaining slopes we searched the orbit objective of Theorem 4 by weighted breakout local search over all shift vectors of order n (96 of them at $n = 12$). At $n = 11$ solutions are found immediately; at $n = 12$ every slope stalls with at least nine violated orbits. Combined with Proposition 6, this is evidence — not a proof — that no shift-orbit construction gives a box-free 2-colouring of $[12]^3$.

Theorem 7 (Dimension raising)**.** *If f is a box-free 2-colouring of $[n]^d$, then*

$$F(x_1, \ldots, x_{d+1}) = f\big((x_1 - x_{d+1}) \bmod n, \ldots, (x_d - x_{d+1}) \bmod n\big)$$

is a box-free 2-colouring of $[n]^{d+1}$. Hence $N(2, d+1) \leq N(2, d)$, and $N(2, d) \geq 12$ for every $d \geq 3$.

Proof. Let B be a box in $[n]^{d+1}$ and split its 2^{d+1} points by the last coordinate into two d-boxes B_u, B_v (the same d-box at the two values of x_{d+1}). If B is monochromatic then B_u and B_v are monochromatic in the same colour. But $F(\cdot, u)$ and $F(\cdot, v)$ are f shifted by u resp. v in every coordinate, and being monochromatic is invariant under a shift of all coordinates, so B_u being monochromatic already means f has a monochromatic d-box, a contradiction. □

Remark 8. The lift was verified exhaustively at $d = 4$: all $\binom{11}{2}^4 = 9\,150\,625$ four-boxes of the lift of the colouring (2) were enumerated, none monochromatic. A sensitivity check (forcing a monochromatic 2^4 sub-block) makes the checker report exactly one box, so the check is not vacuous.

4.1 Reduction to one-dimensional shifts

The colourings of Section 3 are produced by search, not by a formula, and the one algebraic colouring we know explicitly (the quadratic-residue colouring of $n = 11$, Appendix A) is of *sum type*: its colour is a function of $x + y + z$. Motivated by that we consider the family of colourings that are constant on the level sets of $x + y$, i.e. $\chi(x, y, z) = f(x + y, z)$, and show that within this family the three-dimensional box condition collapses to a two-dimensional one.

Fix n and a function f on the set below, and set $\chi_f(x, y, z) = f(x + y, z)$ for $(x, y, z) \in [n]^3$. For a colour γ and two levels $z_1 \neq z_2$ put

$$D_\gamma(z_1, z_2) \;=\; \big\{u \in [0, 2n-2] \;:\; f(u, z_1) = f(u, z_2) = \gamma\big\}, \qquad f\colon [0, 2n-2] \times [n] \to \{0, 1\}.$$

Proposition 9 (Shift criterion)**.** *χ_f is box-free if and only if for every colour γ, every $z_1 < z_2$, every $1 \le a, b \le n-1$ and every $u_0 \in [0,\, 2n-2-a-b]$, the four values $u_0,\ u_0+a,\ u_0+b,\ u_0+a+b$ are* not *all contained in $D_\gamma(z_1, z_2)$.*

Proof. Let $x_1 < x_2$, $y_1 < y_2$, $z_1 < z_2$ and put $b = x_2 - x_1 \ge 1$, $a = y_2 - y_1 \ge 1$ and $u_0 = x_1 + y_1$. The first coordinate $x + y$ of the eight vertices of the box takes exactly the values u_0, $u_0 + a$, $u_0 + b$, $u_0 + a + b$, and the third coordinate takes the two values z_1, z_2. Hence the eight vertices all have colour γ if and only if those four values all lie in $D_\gamma(z_1, z_2)$. (Values may coincide among the four; then some of the eight vertices coincide, but the box is still a box and the four values are still required to lie in D_γ.) The bounds $a, b \le n - 1$ are exactly the range of the differences $y_2 - y_1$ and $x_2 - x_1$. Conversely, let γ, $z_1 < z_2$, $1 \le a, b \le n - 1$ and $u_0 \in [0, 2n - 2 - a - b]$ be such that the four values lie in $D_\gamma(z_1, z_2)$. Since x_1 ranges over $[0, n-1-b]$ and y_1 over $[0, n-1-a]$, the sum $x_1 + y_1$ ranges over all of $[0,\, 2n-2-a-b]$; choose x_1, y_1 with $x_1 + y_1 = u_0$ and put $x_2 = x_1 + b$, $y_2 = y_1 + a$. The eight points then lie in $[n]^3$ and all have colour γ. □

Corollary 10. *For every $z_1 < z_2$ and colour γ the set $D_\gamma(z_1, z_2) \subseteq [2, 2n]$ is free of three-term arithmetic progressions (take $a = b$ in Proposition 9) and free of corners $\{u_0, u_0+a, u_0+b, u_0+a+b\}$ with $a, b \ge 1$.*

Two consequences are worth recording. First, the sum-type colouring $\chi(x, y, z) = g(x + y + z)$ is the special case $f(u, z) = g(u + z)$ (indices reduced modulo n), so Proposition 9 is the natural common framework for the algebraic colourings. Second, the proposition replaces the n^3 free bits of a general colouring by the $2n^2$ bits of f, and turns each slice pair into an independent corner-freeness condition on a finite set — a shape that local search can handle far better than the original constraint. We exploited this in the searches reported in Section 4.2 and we stress that the reduction is *exact*: it neither loses nor gains colourings, so a box-free colouring found in this family is a genuine one.

Remark 11. Corollary 10 should not be read as an impossibility result. It is tempting to conclude that the family dies at $n = 9$ because a two-colouring of $[n]$ always contains a monochromatic three-term progression ($W(2, 3) = 9$); but D_0 and D_1 need not cover $[2, 2n]$ (a value u with $f(u, z_1) \neq f(u, z_2)$ belongs to neither), so no such conclusion follows. We record this because an earlier version of our notes contained the erroneous claim that the family is empty for $n \ge 9$.

4.2 Search in the shift family

Proposition 9 is a constraint satisfaction problem in $2n^2$ bits ($2n - 1$ values of u times n values of z), with one forbidden configuration per colour, per pair $z_1 < z_2$, per $1 \le a, b \le n - 1$ and per admissible u_0. Each forbidden configuration is a clause over eight variables, so the instance has $O(n^5)$ clauses of width eight and only $2n^2$ variables; for $n = 14$ this is 378 variables and 399 854 clauses, against $14^3 = 2744$ variables and 1.5×10^6 clauses for the direct encoding of the original problem. We solved these instances with a CDCL solver (CaDiCaL, via `python-sat`).

Two checks fix the meaning of the results. First, the sum-type colouring $\chi(x,y,z) = g(x+y+z)$ is the case $f(u,z) = g(u+z)$; for $n = 11$ with g the quadratic residues mod 11 the criterion evaluates to zero violations, and the solver re-finds a box-free colouring of $[11]^3$ inside the family in 53 s. Second, every colouring returned is re-verified from the definition by an independent checker (`core.is_valid_fast`), which scans all $\binom{n}{2}^3$ boxes. The family therefore neither loses nor manufactures colourings, as claimed in Section 4.1.

Remark 12. The bound $b \le n-1$ in Proposition 9 is not cosmetic. An earlier version of the criterion imposed only $a, b \ge 1$, which is *strictly stronger*: for $n = 11$ it assigns 360 violations to the genuine colouring of Appendix A and thereby rejects it, because it admits configurations such as $a = 1$, $b = 11$ whose "difference in x" exceeds the width of the grid. We mention this because the failure is silent — the criterion remains sufficient, so a search based on it returns only valid colourings and the error is invisible unless one tests it against a known example.

For local search the useful identity is the following. For a set D of positions and integers $a, b \ge 1$, the number of violating configurations with those a, b is $|D \cap (D-a) \cap (D-b) \cap (D-a-b)|$, and summing over *all* $b \ge 1$ gives $\binom{|D \cap (D-a)|}{2}$, so the unbounded violation count is computable with $O(n)$ bit operations. With the truncation $b \le n-1$ a direct double loop over (a, b) is needed, but each D is then a bitmask and the whole cost of a state is a few thousand popcount operations.

5 Upper bounds

5.1 A Zarankiewicz argument

The following is elementary and self-contained.

Theorem 13. *Let $c \ge 2$, $n \ge 2$ and suppose that*

$$n^2 \binom{n/c}{2} \;>\; \binom{n}{2} Z_2(n), \qquad Z_2(n) := \tfrac{n}{2} + \tfrac{n}{2}\sqrt{4n-3}. \tag{4}$$

Then $[n]^3$ is c-guaranteed.

Proof. Let $f\colon [n]^3 \to [c]$ be box-free. For a colour γ and two slices $x_1 < x_2$ put $B_\gamma(x_1, x_2) = \{(y,z) : f(x_1,y,z) = f(x_2,y,z) = \gamma\} \subseteq [n]^2$. If $B_\gamma(x_1,x_2)$ contained a rectangle, its eight points together with x_1, x_2 would form a monochromatic box; so $B_\gamma(x_1,x_2)$ is C_4-free as a bipartite graph and therefore has at most $Z_2(n)$ elements, since a C_4-free bipartite graph with parts of size n and e edges satisfies $\sum_y \binom{d_y}{2} \le \binom{n}{2}$ and, by convexity, $\sum_y \binom{d_y}{2} \ge (e^2/n - e)/2$.

Write $m_\gamma(y,z) = \#\{x : f(x,y,z) = \gamma\}$. Double counting gives $\sum_{x_1<x_2} |B_\gamma(x_1,x_2)| = \sum_{y,z} \binom{m_\gamma(y,z)}{2}$. Since $\sum_\gamma m_\gamma(y,z) = n$ and $t \mapsto \binom{t}{2}$ is convex, $\sum_\gamma \binom{m_\gamma(y,z)}{2} \ge c\binom{n/c}{2}$ for every (y,z). Hence

$$n^2\, c\binom{n/c}{2} \;\le\; \sum_{\gamma} \sum_{x_1<x_2} |B_\gamma(x_1,x_2)| \;\le\; \binom{n}{2} c\, Z_2(n). \qquad \square$$

Numerically (4) first fails at $n = 22$ for $c = 2$, giving $N(c,d)[3] \le 22$; the same computation gives 93, 277, 657, 1341 for $c = 3, 4, 5, 6$.

5.2 A packing bound and an improvement for $c = 2$

In the feature language of Section **??**, Lemma 15 of [2] says that the n slices of a box-free colouring must use *pairwise disjoint* features. Since a slice uses exactly as many features as it has monochromatic rectangles,

$$n\,\varphi_c(n) \;\le\; c\binom{n}{2}^2 \tag{5}$$

is necessary for a box-free c-colouring of $[n]^3$ to exist, where $\varphi_c(n)$ is the least number of monochromatic rectangles in a c-colouring of $[n]^2$. Cooper–Fenner–Purewal bound φ_c through their recursion. For $c = 2$ we bound $\varphi_2(n)$ by two exact levels of convexity, and the integrality of the intermediate composition decides $n = 21$.

Theorem 14. $\varphi_2(21) \geq 4205$. *Consequently* $[21]^3$ *is* 2*-guaranteed, i.e.* $N(c,d)[3] \leq 21$ *for* $c = 2$.

Proof. Let k_y be the number of 0-cells in row y of a 2-colouring of $[21]^2$. For each of the $\binom{21}{2} = 210$ column pairs j let a_j (resp. b_j) be the number of rows whose two cells in j are both 0 (resp. both 1). The number of monochromatic rectangles is

$$R = \sum_j \Big[\binom{a_j}{2} + \binom{b_j}{2}\Big], \qquad \sum_j a_j = T_0 := \sum_y \binom{k_y}{2}, \quad \sum_j b_j = T_1 := \sum_y \binom{21-k_y}{2}.$$

For integers $x_j \geq 0$ with $\sum_j x_j = T$ convexity gives $\sum_j \binom{x_j}{2} \geq \mu(T, 210)$, where μ is attained by distributing T as evenly as possible over the 210 bins; hence $R \geq \mu(T_0, 210) + \mu(T_1, 210)$. Each row contributes $\binom{k}{2} + \binom{21-k}{2} \geq \binom{10}{2} + \binom{11}{2} = 100$, an even number. Therefore either $T_0 + T_1 = 2100$, in which case every row has $k_y \in \{10, 11\}$ and $(T_0, T_1) = (945 + 10m,\ 1155 - 10m)$ for some integer $0 \leq m \leq 21$; or $T_0 + T_1 \geq 2102$. In the first case $\mu(T_0, 210) + \mu(T_1, 210)$ is minimised at $m = 10, 11$, where it equals $\mu(1045, 210) + \mu(1055, 210) = 2080 + 2125 = 4205$; in the second it is at least $2\mu(1051, 210) = 4210$. Hence $R \geq 4205$. Finally (5) at $c = 2$, $n = 21$ would require $21 \cdot 4205 = 88305 \leq 2\binom{21}{2}^2 = 88200$, which fails. □

Remark 15. The obstruction is a projective plane. The continuous form of the argument ($T_0 = T_1 = 1050$) gives only $\varphi_2(21) \geq 4200$ and $21 \cdot 4200 = 88200$, *exactly* the number of features available; $21 = 4^2+4+1$ is a projective-plane order and the extremal Zarankiewicz value $Z_2(21) = 105 = 5 \cdot 21$ is attained by the plane of order 4, which is also the extremal example for the sharpness of the Kővari–Sós–Turán bound in the case $t = 2$. So no refinement of the continuous count can pass $n = 21$: the improvement above comes precisely from the fact that $T_0 = T_1 = 1050$ is not attainable by an integer row composition, and the least integer excess costs 105 features.

Remark 16. Applying the same computation with c colours, (5) first fails at $n = 93,277,657$ for $c = 3, 4, 5$ — the same thresholds as Lemma 4 of [2], so Theorem 14 is an improvement only for $c = 2$.

5.3 Relation to the literature

The Zarankiewicz Theorem 13 is *not* an improvement: it reproduces Lemma 4 of [2] applied to the cube (the values $22, 93, 277, 657$ for $c = 2, 3, 4, 5$). Theorem 14 improves the $c = 2$ value from 22 to 21; we found no post-2008 improvement of the cube bounds in the literature, and [2] itself remarks that "the case of $c = 2$ and $d = 3$" is where the interesting computational questions remain open.

5.4 Resulting windows

Combining the constructions of Section 3 with the upper bounds, and using the published consequence $N(2, 3) \geq 13$ (see the next paragraph):

The $c = 2$ lower bound deserves a word. Our own best construction gives $[11]^3$, i.e. $N(2, 3) \geq 12$, which is *weaker* than what is already implied by [2]: their table records that $[12, 12, 27]$ is 2-colourable, and restricting such a colouring to the subgrid $[12]^3$ gives $N(2, 3) \geq 13$. Thus

$$13 \leq N(2, 3) \leq 21,$$

The lower bound is not due to us. The upper bound is Theorem 14; the previous best value, 22, follows from Lemma 4 of [2] (and is reproduced by our Zarankiewicz argument), while the closed-form corollary of [2] gives only 80. We therefore regard $[13, 21]$ as the concrete outcome for $c = 2$, with the improvement over [2] being the single unit obtained from Remark 15.

c	lower bound	upper bound	status
1	2	2	exact
2	13 (implied by [2, Fig. 3])	**21** (Thm. 14)	window of width 8
3	24 (this note)	93	window of width 70
4	41 (this note)	277	window of width 237
5	45 (this note)	657	window of width 613

Table 3: Two-sided windows for $N(c,d)[3]$.

6 The growth of $N(2,d)$

Theorem 17. *For every $d \geq 2$ and every n with $\binom{n}{2}^d < 2^{2^{d-1}}$ there is a box-free 2-colouring of $[n]^d$. In particular*

$$N(2,d) \;>\; 2^{2^d/(2d)}.$$

Proof. Colour the n^d points independently and uniformly. A fixed box is monochromatic in either colour with probability 2^{1-2^d}, so the union bound gives $\Pr[\exists \text{ monochromatic box}] \leq \binom{n}{2}^d 2^{1-2^d} < 1$. Using $\binom{n}{2} \geq n^2/4$ yields $n^{2d}/4^d < 2^{2^d-1}$, i.e. $n < 2^{(2^d-1)/(2d)+1}$, which implies the claim for $d \geq 2$. □

The question whether $N(2,d) \geq 2^{c\,2^d}$ holds for a *fixed* $c > 0$ is open; it would amount to improving the exponent by a factor d. Four independent counting arguments all produce the same $2^d/d$ rate:

(a) the union bound above ($2^d/(2d)$);

(b) the Lovász local lemma, with dependency degree at most $2^d \prod(a_i - 1)$ [2, Thm. 2], again $2^d/(2d)$;

(c) the layer count of Proposition 2: n slices each using ν features out of $F = 2\binom{n}{2}^{d-1}$ satisfy the birthday-type condition $n\nu^2 \lesssim F$, which for random-quality slices gives $n^{2d-1} \lesssim 2^{2^d}$, i.e. $2^d/(2d-1)$;

(d) the Zarankiewicz argument of Theorem 13 in general dimension, which asks for the largest box-free *subset* of $[n]^d$ of density $1/c^2$ and again gives $2^d/d$ in the exponent.

Routes (c) and (d) indicate exactly what would be needed: a family of colourings (or of box-free subsets) whose number of monochromatic boxes is a factor $2^{\Theta(2^d)}$ below the random count. Direct measurement suggests this does not happen: minimising the number of monochromatic boxes over 2-colourings of $[n]^m$ by local search gives 255 versus a random expectation of 756 at $(m,n) = (2,11)$, 424 versus 1089 at $(2,12)$, and 65 versus 364 at $(3,9)$ — ratios $0.34, 0.39, 0.18$, decreasing slowly in m but far from exponential. We therefore conjecture that the factor d in the exponent is not an artefact of the known arguments, and that a uniform c requires a different idea.

7 Discussion and open problems

(1) **Exact values.** Not a single $N(c,d)[3]$ with $c \geq 2$ is known. The most accessible target is $N(2,3) \in [13,21]$. On the lower side one needs a box-free 2-colouring of some $[n]^3$ with $13 \leq n \leq 20$. On the upper side Theorem 14 shows that the binding quantity is $\varphi_2(n)$, the least number of monochromatic rectangles in a 2-colouring of $[n]^2$: by (5) any improvement of the lower bound $\varphi_2(n) \geq \mu(T_0, \binom{n}{2}) + \mu(T_1, \binom{n}{2})$ that pushes $n\varphi_2(n)$ past $2\binom{n}{2}^2$ for some

smaller n immediately lowers $N(2,3)$. Our convexity bound is loose by roughly a factor of two already at $n = 5$ (1 against the exact value 2, which we verified by exhaustive enumeration of column multisets), so there is room. The *threshold* version of φ_2 is classical: $\varphi_2(n) = 0$ for $n \le 4$ and $\varphi_2(n) \ge 1$ for $n \ge 5$, the latter being the bipartite Ramsey number $r(C_4, C_4) = 5$. What is not, to our knowledge, tabulated is the multiplicity $\varphi_2(n)$ itself for $n \ge 6$. Determining it exactly is a self-contained two-dimensional problem — the bipartite analogue of the Ramsey multiplicity problem — and it is the quantity that Theorem 14 makes binding.

(2) **Beyond the shift family.** Proposition 9 and Section 4.2 reduce the three-dimensional problem, for colourings constant on the level sets of $x + y$, to a two-dimensional corner condition. The natural next step is the family of colourings constant on the level sets of a general linear form, $\chi(x, y, z) = f(\alpha x + \beta y, z)$, or of two independent forms; each such choice yields its own clause set and its own notion of forbidden configuration, and we have not explored whether one of them is large enough to contain a colouring with $n \ge 13$.

(3) **Permutation families.** Replacing the cyclic shifts in (3) by arbitrary per-slice permutations π_x of $[n]$ gives a family to which the orbit argument does not apply. A pigeonhole estimate shows that a random permutation family fails (for $n = 12$ the expected number of collisions per constraint is about 2.7), so a successful design would have to keep $|S(\gamma, \{z_1, z_2\})| \le 3$ throughout; this is a $\lambda \le 3$ design constraint and is, in our view, the most promising route to $n = 12$ or beyond.

(4) **The exponent gap.** Whether $N(2, d) \ge 2^{c\,2^d}$ holds for a fixed $c > 0$ is open. Sections 6 isolates the equivalent box-multiplicity question.

Contributions

This work was carried out jointly by the author and an AI agent (Arena.ai Agent Mode), and the division of labour was as follows.

The author (Dmitry Kamenetsky) posed the problem and fixed its scope: the object studied (axis-aligned combinatorial boxes / prisms with eight vertices), the range $c \le 5$, and the sequence of goals pursued — from producing explicit colourings, to attacking $n = 12$, to the growth question and the upper-bound side. The author directed the investigation at each stage, specified the deliverables, and reviewed all intermediate results; in particular the author's decision to ask whether the obtained bounds were actually improvements led to the literature comparison in Section 5 and to the withdrawal of the upper-bound claim. The author is responsible for the framing, the selection of what to report, and the final text.

The AI agent wrote essentially all of the software: the brute-force and slice-pair verifiers, the SAT and CP-SAT encodings, the layer-packing constructor with its bit-mask local search, the local-search and portfolio searches for the shift families, and the numerics for the scaling and Zarankiewicz computations. It ran the experiments and produced the colourings tabulated in Table 1. It proposed and proved the structural results of Section 4 (the orbit criterion, the Sidon analysis of the single-coordinate family, the dimension-raising lemma), the C_4 argument of Section 5, and the observations of Section 6, and drafted the present text and its LaTeX source. It also carried out the literature comparison that showed that the Zarankiewicz upper bounds coincide with Lemma 4 of [2], that the $c = 2$ lower bound is implied by [2, Fig. 3], and that the earlier draft's claim of $N(2,3) \le 22$ as a new result was wrong; the improvement to 21 (Theorem 14) was found by the AI agent, which also wrote the standalone certificate `code/final_bound.py` for it; that certificate is short enough to be checked by hand and is the

artifact the author should verify before submission. The AI agent also proposed the shift reduction of Section 4.1 and its search form, and — more usefully — the record of one failure: the first version of Proposition 9 omitted the bound $a, b \le n-1$, which made the criterion strictly stronger than necessary; the error was detected only because the criterion was tested against the known $n = 11$ colouring, which it wrongly rejected (Remark after Proposition 9). Every computational claim was checked by two independent verifiers as described in Section 3, and all proofs are written out in full so that they can be checked by a human reader.

References

[1] G. Agnarsson, B. Doerr, T. Schoen, *Coloring t-dimensional m-boxes*, Discrete Math. **226** (2001), 21–33.

[2] J. Cooper, S. Fenner, S. Purewal, *Monochromatic boxes in colored grids*, arXiv:0810.3019 (2008); also available from the authors' web pages.

[3] S. Fenner, W. Gasarch, C. Glover, S. Purewal, *Rectangle free coloring of grids*, arXiv:1005.3750.

A Colourings

We record our best construction for $c = 2$, i.e. a box-free 2-colouring of $[11]^3$, together with the rule that generates it. All colourings, for every (c, n) of Table 1, are stored in the accompanying repository as JSON arrays `grid[x][y][z]` (zero-based).

The $c = 2$, $n = 11$ colouring. Let $Q = \{1, 3, 4, 5, 9\} \subset \mathbb{Z}_{11}$ be the set of non-zero quadratic residues modulo 11 (equivalently, the base block of the $(11, 5, 2)$ symmetric difference set), and for $y \in \{0, \dots, 10\}$ put $T_y = (Q + y) \bmod 11$. Define

$$f(x, y, z) = 1 \iff (z - x) \bmod 11 \in T_y.$$

Explicitly,

$$\begin{array}{ll} T_0 = \{0, 1, 2, 4, 5, 7\} & T_1 = \{0, 1, 3, 5, 6, 10\} \\ T_2 = \{0, 2, 3, 9, 10\} & T_3 = \{0, 3, 5, 7, 8, 9\} \\ T_4 = \{0, 4, 6, 8, 9\} & T_5 = \{1, 2, 3, 6, 8\} \\ T_6 = \{1, 4, 5, 8, 9, 10\} & T_7 = \{1, 6, 7, 9, 10\} \\ T_8 = \{2, 3, 4, 5, 6, 9\} & T_9 = \{2, 5, 6, 7, 8, 10\} \\ T_{10} = \{3, 4, 7, 8, 10\} & \end{array}$$

Why it works: for $y_1 \neq y_2$ the translate structure gives $|T_{y_1} \cap T_{y_2}| = 2$ whenever $y_1 - y_2$ is a non-zero residue and $|T_{y_1} \cap T_{y_2}| = 0$ when $y_1 - y_2$ is a non-residue, the two common columns always differing by ± 2; the complementary intersections are translates of Q itself. Each pair of rows therefore contributes one difference class only, which is precisely the Sidon-type condition of Proposition 6, so the colouring is box-free. An exhaustive check of all $\binom{11}{2}^3 = 166\,375$ boxes confirms this.

Other sizes. The repository contains box-free colourings for $c = 2$ with $n \in \{9, 10, 11\}$; for $c = 3$ with $n \in \{11, 12, 13, 14, 15, 17, 20, 22, 23\}$; for $c = 4$ with $n \in \{11, 14, 16, 18, 20, 22, 24, 28, 32, 36, 40\}$, and for $c = 5$ with $n \in \{11, 24, 28, 32, 36, 40, 44\}$. The largest in each case is the one quoted in Table 1.